CoRoT-2b

R-1698

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_130906` · created 2026-08-02T13:55:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456541.776 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.014
Transit depth (Rp/Rs)^2	2.91 +/- 0.49 [%]
Orbital Inclination (inc)	87.62 +/- 1.28
Airmass coefficient 1 (a1)	1.015 +/- 0.017
Airmass coefficient 2 (a2)	-0.0063 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	1.65 %
Best Comparison Star	#3 - [243, 301]
Optimal Aperture	3.06
Optimal Annulus	14.83
Transit Duration (day)	0.0939 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.014 vs 0.1667 (deviation 0.28σ)
Duration fit vs published	0.0939 d vs 0.095 d (deviation 0.3σ)
Mid-time O–C	-3.6 min
Depth SNR	11.2
Residual scatter	1.65 %

Light curve

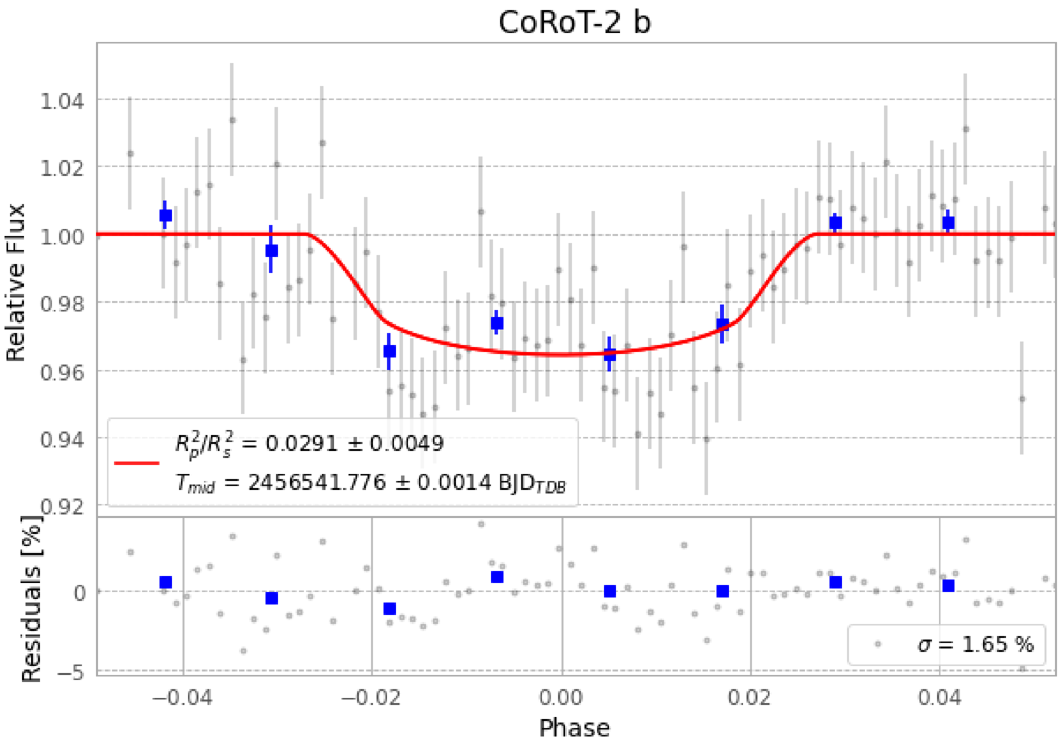

Figure 1 — FinalLightCurve_CoRoT-2 b_2013-09-05.png (SHA-256 57334bc6e10af422…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [333, 277], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a66c24df27be2097…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

85 FITS frames (56 MB), `CoRoT-2130906042715.FITS` → `CoRoT-2130906084212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-130906.log (a0d836925e78…), FinalLightCurve_CoRoT-2 b_2013-09-05.png (57334bc6e10a…), FinalLightCurve_CoRoT-2 b_2013-09-05.csv (33e96aa348ed…)

Compute resources

Wall time 165.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)